American International University-Bangladesh (AIUB)

Department of computer science
Faculty of Science & Technology (FST)

Semester: FALL 2025-26

Course: Introduction to Data Science

Section : J

Group: A

Project Title:

The Global Crisis Monitor:

Exploratory Analysis and Classification of Humanitarian Disasters

Data source: ReliefWeb – Global Humanitarian Disaster Listings

Supervised By:

Kamrun Naher Koli

Lecturer,
Department of Computer Science
Faculty of Science and Technology
American International University-Bangladesh

Submitted By :

Name	ID
Taharat Muhammad Jabir	**22-49037-3**
Merin Taj	**23-50865-1**
Wasimul Bari Rahul	**23-50913-1**
MD.Hasibul Islam	**23-51777-2**

Submission Date : 05-01-2026

1. Introduction

Humanitarian disasters such as floods, earthquakes, epidemics, and tropical cyclones pose serious threats to human life, infrastructure, and socio-economic stability. Monitoring disaster occurrences and understanding their patterns across regions and time are essential for effective disaster preparedness, mitigation, and humanitarian response.

With the increasing availability of real-time humanitarian data, data science techniques can be applied to extract meaningful insights from disaster records. This project utilizes global disaster data obtained from ReliefWeb, a humanitarian information platform managed by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA). The dataset contains structured disaster listings from multiple countries and across several decades.

The study follows a complete data science pipeline that includes web scraping, data cleaning, exploratory data analysis (EDA), and supervised machine learning classification. The analysis aims to provide insights into disaster distribution by type, geography, and time, as well as evaluate the feasibility of predicting disaster categories using historical attributes.

2. Project Objective

The main objectives of this project are:

1. To collect real-world humanitarian disaster data from the ReliefWeb website using web scraping techniques
2. To clean and preprocess the scraped data for analysis
3. To perform exploratory data analysis (EDA) to identify patterns and trends in disaster occurrences
4. To visualize disaster distributions by type, country, and year
5. To build and evaluate a supervised classification model to predict disaster categories
6. To analyze feature importance and model performance for disaster classification

3. Tools and Libraries Used

This project was implemented using the **R programming language** with the following libraries:

Library	Purpose
rvest	Web scraping and HTML parsing
dplyr	Data manipulation and transformation
stringr	String processing and regular expressions
tidyr	Data reshaping and long-format conversion
ggplot2	Data visualization
ranger	Random Forest classification

```r
# Part 1: Libraries
library(rvest)
library(dplyr)
library(ggplot2)
library(stringr)
library(tidyr)
library(ranger)
```

4. Data Collection

4.1 Data Source

The dataset was scraped from the ReliefWeb disaster listings:

https://reliefweb.int/disasters

ReliefWeb is a global humanitarian information service operated by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA). The platform serves as a centralized repository for information related to humanitarian crises, including natural disasters, conflicts, and complex emergencies worldwide. It aggregates data from governments, international organizations, non-governmental organizations (NGOs), and humanitarian agencies.

The ReliefWeb disaster listings provide structured and regularly updated records of disaster events, including information such as disaster type, affected countries, disaster status (alert, ongoing, or past), and event timelines. The website prioritizes recent and ongoing disasters, making it particularly useful for real-time monitoring and analysis.

Due to its authoritative sources, global coverage, and standardized reporting format, ReliefWeb is widely used by humanitarian practitioners, researchers, and policy makers. However, as the platform is designed primarily for operational awareness rather than data analytics, the data requires preprocessing and cleaning before being used for statistical analysis or machine learning tasks.

4.2 Web Scraping Methodology

- A total of **184 pages** (pages 0–183) were scraped to capture historical records.
- Disaster information was extracted by inspecting HTML <article> elements.
- Each request was delayed using Sys.sleep(1) to ensure ethical scraping.

```r
# Part 2: Web Scraping
base_url <- "https://reliefweb.int/disasters?page="
TOTAL_PAGES <- 184  # pages 0..183

all_rows <- list()

for (page in 0:(TOTAL_PAGES - 1)) {
  url <- paste0(base_url, page)
  page_df <- tryCatch({

    webpage <- read_html(url)
    cards <- webpage %>%
      html_elements("article.rw-river-article--disaster")

    rows <- lapply(seq_along(cards), function(i) {
      card <- cards[[i]]

      disaster_id <- card %>% html_attr("data-id")
      status_code <- card %>% html_attr("data-disaster-status")
      title_node <- card %>% html_element("h3.rw-river-article__title a")
      title <- title_node %>% html_text() %>% str_trim()
      dtypes <- card %>%
        html_elements("dd.rw-entity-meta__tag-value--disaster-type a.rw-entity-meta__tag-link") %>%
        html_text() %>%
        str_trim()
      disaster_type_raw <- if (length(dtypes) == 0) NA_character_ else paste(unique(dtypes), collapse = ", ")

      countries <- card %>%
        html_elements("dd.rw-entity-meta__tag-value--country a.rw-entity-meta__tag-link") %>%
        html_text() %>%
        str_trim()
      countries_raw <- if (length(countries) == 0) NA_character_ else paste(unique(countries), collapse = ", ")

      data.frame(
        DisasterID = disaster_id,
        Title = title,
        DisasterTypeRaw = disaster_type_raw,
        StatusCode = status_code,
        CountriesRaw = countries_raw,
        stringsAsFactors = FALSE
      )
    })

    bind_rows(rows)

  }, error = function(e) {
    message("Error scraping page ", page, ": ", conditionMessage(e))
    data.frame()
  })

  all_rows[[page + 1]] <- page_df
  Sys.sleep(1)
}
disaster_db <- bind_rows(all_rows) %>%
  filter(!is.na(Title) & Title != "") %>%
  distinct()

cat("\nTotal rows scraped (raw): ", nrow(disaster_db), "\n", sep = "")
```

Output:

```
Total rows scraped (raw): 3663
```

Dataset Structure (Pre-cleaning)

Variable	Description	Type
DisasterID	Unique identifier from ReliefWeb	Character
Title	Disaster title (includes date)	Character
DisasterTypeRaw	Raw disaster category	Character
StatusCode	Disaster status	Character
CountriesRaw	Affected countries	Character

5. Data Cleaning and Preprocessing

5.1 Missing Value Analysis

A comprehensive missing value audit was conducted before cleaning. Both NA values and blank strings were examined for each column.

Observation:

No missing values were found in the raw scraped attributes. However, some rows were removed due to failures in extracting the year from disaster titles.

```r
# Part 3: Missing Value Audit
rows_before <- nrow(disaster_db)

na_counts <- sapply(disaster_db, function(x) sum(is.na(x)))
blank_counts <- sapply(disaster_db, function(x) sum(str_trim(ifelse(is.na(x), "", x)) == ""))

missing_audit <- data.frame(
  Column = names(na_counts),
  NA_Count = as.integer(na_counts),
  BlankString_Count = as.integer(blank_counts),
  TotalMissing = as.integer(na_counts + blank_counts),
  row.names = NULL
)

cat("\n--- Missing Value Audit (NA + Blank Strings) ---\n")
print(missing_audit)

# Part 4: Drop-Reason Audit
month_re <- "(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)"
audit <- disaster_db %>%
  mutate(
    TitleDate = str_extract(Title, paste0(month_re, "\\s\\d{4}$")),
    YearParsed = suppressWarnings(as.integer(str_extract(TitleDate, "\\d{4}"))),

    Missing_DisasterID = is.na(DisasterID) | str_trim(DisasterID) == "",
    Missing_Title = is.na(Title) | str_trim(Title) == "",
    Missing_TypeRaw = is.na(DisasterTypeRaw) | str_trim(DisasterTypeRaw) == "",
    Missing_Status = is.na(StatusCode) | str_trim(StatusCode) == "",
    Missing_Countries = is.na(CountriesRaw) | str_trim(CountriesRaw) == "",
    Failed_YearParse = is.na(YearParsed)
  )
drop_reason_table <- data.frame(
  Reason = c(
    "Missing DisasterID",
    "Missing Title",
    "Missing DisasterTypeRaw (NA/blank)",
    "Missing StatusCode (NA/blank)",
    "Missing CountriesRaw (NA/blank)",
    "Failed Year Parse from Title"
  ),
  Rows = c(
    sum(audit$Missing_DisasterID),
    sum(audit$Missing_Title),
    sum(audit$Missing_TypeRaw),
    sum(audit$Missing_Status),
    sum(audit$Missing_Countries),
    sum(audit$Failed_YearParse)
  ),
  row.names = NULL
)
cat("\n--- Drop-Reason Audit (Pre-cleaning) ---\n")
print(drop_reason_table)
```

Output:

```
--- Missing Value Audit (NA + Blank Strings) ---
> print(missing_audit)
           Column NA_Count BlankString_Count TotalMissing
1      DisasterID        0                 0            0
2           Title        0                 0            0
3 DisasterTypeRaw        0                 0            0
4      StatusCode        0                 0            0
5    CountriesRaw        0                 0            0
```

```
--- Drop-Reason Audit (Pre-cleaning) ---
> print(drop_reason_table)
                              Reason Rows
1                 Missing DisasterID    0
2                      Missing Title    0
3 Missing DisasterTypeRaw (NA/blank)    0
4      Missing StatusCode (NA/blank)    0
5    Missing CountriesRaw (NA/blank)    0
6        Failed Year Parse from Title   50
```

5.2 Feature Engineering

The following preprocessing steps were applied:

- Extracted Year and Month from disaster titles using regular expressions
- Selected the primary disaster type when multiple types were present
- Removed duplicate disaster records using DisasterID
- Assigned a sequential RowID after cleaning

```r
# Part 5: Cleaning & Feature Engineering
clean_disasters <- disaster_db %>%
  mutate(
    TitleDate = str_extract(Title, paste0(month_re, "\\s\\d{4}$")),
    Year = suppressWarnings(as.integer(str_extract(TitleDate, "\\d{4}"))),
    Month = str_extract(TitleDate, paste0("^", month_re)),
    DisasterType = ifelse(
      is.na(DisasterTypeRaw),
      NA_character_,
      str_split(DisasterTypeRaw, ",\\s*") %>% sapply(`[`, 1)
    )
  ) %>%
  filter(
    !is.na(DisasterID) & str_trim(DisasterID) != "",
    !is.na(Title) & str_trim(Title) != "",
    !is.na(Year),
    !is.na(DisasterType) & str_trim(DisasterType) != "",
    !is.na(StatusCode) & str_trim(StatusCode) != "",
    !is.na(CountriesRaw) & str_trim(CountriesRaw) != ""
  ) %>%
  distinct(DisasterID, .keep_all = TRUE) %>%
  mutate(RowID = row_number()) %>%
  relocate(RowID)

rows_after <- nrow(clean_disasters)
rows_removed <- rows_before - rows_after

cat("\nRows before cleaning:", rows_before, "\n")
cat("Rows after cleaning :", rows_after, "\n")
cat("Rows removed        :", rows_removed, "\n")
```

Output:

```
Rows before cleaning: 3663
> cat("Rows after cleaning :", rows_after, "\n")
Rows after cleaning : 3613
> cat("Rows removed        :", rows_removed, "\n")
Rows removed        : 50
>
```

6. Dataset Overview

To verify data integrity and ordering, the dataset was inspected by displaying:

- The first 15 records
- The last 15 records
- A random sample of 15 records

```r
# Part 6: First 15 / Last 15 / Random 15
first_15 <- clean_disasters %>% slice_head(n = 15)
last_15  <- clean_disasters %>% slice_tail(n = 15)

set.seed(42)
random_15 <- clean_disasters %>% sample_n(15)

cat("\n--- FIRST 15 ROWS ---\n"); print(first_15)
cat("\n--- LAST 15 ROWS ---\n");  print(last_15)
cat("\n--- RANDOM 15 ROWS ---\n"); print(random_15)
```

Output:

```
--- FIRST 15 ROWS ---
   RowID DisasterID                                      Title  DisasterTypeRaw StatusCode CountriesRaw TitleDate Year
1      1      52495                      Iran: Floods - Dec 2025      Flash Flood    ongoing         Iran  Dec 2025 2025
2      2      52491 occupied Palestinian territory: Floods - Nov 2025        Flood      alert          oPt  Nov 2025 2025
3      3      52497                   Morocco: Floods - Dec 2025            Flood      alert      Morocco  Dec 2025 2025
4      4      52494                      Iraq: Floods - Dec 2025      Flash Flood    ongoing         Iraq  Dec 2025 2025
5      5      52482            Tropical Cyclone Ditwah - Nov 2025            Flood    ongoing    Sri Lanka  Nov 2025 2025
6      6      52484                  Malaysia: Floods - Nov 2025            Flood    ongoing     Malaysia  Nov 2025 2025
7      7      52489                     Congo: Floods - Nov 2025            Flood    ongoing        Congo  Nov 2025 2025
8      8      52478   Ethiopia: Marburg Disease Outbreak - Nov 2025      Epidemic    ongoing     Ethiopia  Nov 2025 2025
9      9      52483    Indonesia: Floods and Landslides - Nov 2025          Flood    ongoing    Indonesia  Nov 2025 2025
10    10      52481                  Thailand: Floods - Nov 2025      Flash Flood    ongoing     Thailand  Nov 2025 2025
11    11      52474                 Typhoon Fung-Wong - Nov 2025 Tropical Cyclone    ongoing  Philippines  Nov 2025 2025
12    12      52467             Afghanistan: Earthquake - Nov 2025     Earthquake    ongoing  Afghanistan  Nov 2025 2025
13    13      52461                  Hurricane Melissa - Oct 2025            Flood    ongoing        Haiti  Oct 2025 2025
14    14      52462           Cuba: Arboviral Outbreak - Oct 2025         Epidemic    ongoing         Cuba  Oct 2025 2025
15    15      52458                     Egypt: Floods - Oct 2025            Flood    ongoing        Egypt  Oct 2025 2025
   Month     DisasterType
1    Dec      Flash Flood
2    Nov            Flood
3    Dec            Flood
4    Dec      Flash Flood
5    Nov            Flood
6    Nov            Flood
7    Nov            Flood
8    Nov         Epidemic
9    Nov            Flood
10   Nov      Flash Flood
11   Nov Tropical Cyclone
12   Nov       Earthquake
13   Oct            Flood
14   Oct         Epidemic
15   Oct            Flood
```

```
--- LAST 15 ROWS ---
   RowID DisasterID                                   Title  DisasterTypeRaw StatusCode CountriesRaw TitleDate Year
1   3599       6243                Bangladesh: Floods - May 1984          Flood       past   Bangladesh  May 1984 1984
2   3600       5588                    Myanmar: Fire - Apr 1984       Wild Fire       past      Myanmar  Apr 1984 1984
3   3601       6693                Ethiopia: Drought - Apr 1984         Drought       past     Ethiopia  Apr 1984 1984
4   3602       5490                Botswana: Drought - Apr 1984         Drought       past     Botswana  Apr 1984 1984
5   3603       4616 Papua New Guinea: Rabaul Volcano Alert - Apr 1984   Volcano       past          PNG  Apr 1984 1984
6   3604       6702                    Myanmar: Fire - Mar 1984       Wild Fire       past      Myanmar  Mar 1984 1984
7   3605       4835      Indonesia: Floods and Landslides - Mar 1984      Flood       past    Indonesia  Mar 1984 1984
8   3606       5997               Madagascar: Cyclone - Feb 1984 Tropical Cyclone     past   Madagascar  Feb 1984 1984
9   3607       6439                   Benin: Drought - Feb 1984         Drought       past        Benin  Feb 1984 1984
10  3608       6271                Swaziland: Floods - Feb 1984           Flood       past     Eswatini  Feb 1984 1984
11  3609       6102              Mauritania: Drought - Dec 1983         Drought       past   Mauritania  Dec 1983 1983
12  3610       5752                 Bolivia: Drought - Aug 1983         Drought       past      Bolivia  Aug 1983 1983
13  3611       5872           El Salvador: Earthquake - Jun 1982     Earthquake       past  El Salvador  Jun 1982 1982
14  3612       5925         Mozambique: Drought/Floods - Mar 1982       Drought       past   Mozambique  Mar 1982 1982
15  3613       5060        Chad: Drought/Civil Strife - Nov 1981        Drought       past         Chad  Nov 1981 1981
   Month     DisasterType
1    May            Flood
2    Apr        Wild Fire
3    Apr          Drought
4    Apr          Drought
5    Apr          Volcano
6    Mar        Wild Fire
7    Mar            Flood
8    Feb Tropical Cyclone
9    Feb          Drought
10   Feb            Flood
11   Dec          Drought
12   Aug          Drought
13   Jun       Earthquake
14   Mar          Drought
15   Nov          Drought
```

```
--- RANDOM 15 ROWS ---
   RowID DisasterID                                                          Title        DisasterTypeRaw StatusCode
1   2609       4634                    Nigeria: Ammunition Dump Explosion - Jan 2002 Technological Disaster       past
2   2369       5084 East and South Asia: Typhoon Aere and Typhoon Chaba - Aug 2004       Tropical Cyclone       past
3   1177      12627                                   Namibia: Drought - May 2013                Drought       past
4   1098      14139                        Indonesia: Mt. Kelud Volcano - Feb 2014                Volcano       past
5   1252      11338                        Paraguay: Severe Local Storm - Sep 2012      Severe Local Storm       past
6   3218       4643                          Zimbabwe: Cholera Epidemic - Feb 1993               Epidemic       past
7    634      49193                                     Senegal: Floods - Sep 2019            Flash Flood       past
8   2097       6190                                   Argentina: Floods - Mar 2007                  Flood       past
9   1152      13202                            Lao PDR: Dengue Outbreak - Aug 2013               Epidemic       past
10  1327      10426                              Yemen: Measles Outbreak - Mar 2012               Epidemic       past
11  2072       6668                Afghanistan: Flash Floods and Landslide - Jun 2007            Flash Flood       past
12   356      51369                                    Cameroon: Floods - Aug 2022                  Flood       past
13  1625       6461                        Pakistan: Landslides and Floods - Jan 2010                  Flood       past
14   165      51992                                       Yemen: Floods - Apr 2024                  Flood       past
15  2670       5231                                     Hungary: Floods - Mar 2001                  Flood       past
   CountriesRaw TitleDate Year Month           DisasterType
1       Nigeria  Jan 2002 2002   Jan Technological Disaster
2         China  Aug 2004 2004   Aug       Tropical Cyclone
3       Namibia  May 2013 2013   May                Drought
4     Indonesia  Feb 2014 2014   Feb                Volcano
5      Paraguay  Sep 2012 2012   Sep     Severe Local Storm
6      Zimbabwe  Feb 1993 1993   Feb               Epidemic
7       Senegal  Sep 2019 2019   Sep            Flash Flood
8     Argentina  Mar 2007 2007   Mar                  Flood
9       Lao PDR  Aug 2013 2013   Aug               Epidemic
10        Yemen  Mar 2012 2012   Mar               Epidemic
11  Afghanistan  Jun 2007 2007   Jun            Flash Flood
12     Cameroon  Aug 2022 2022   Aug                  Flood
13     Pakistan  Jan 2010 2010   Jan                  Flood
14        Yemen  Apr 2024 2024   Apr                  Flood
15      Hungary  Mar 2001 2001   Mar                  Flood
```

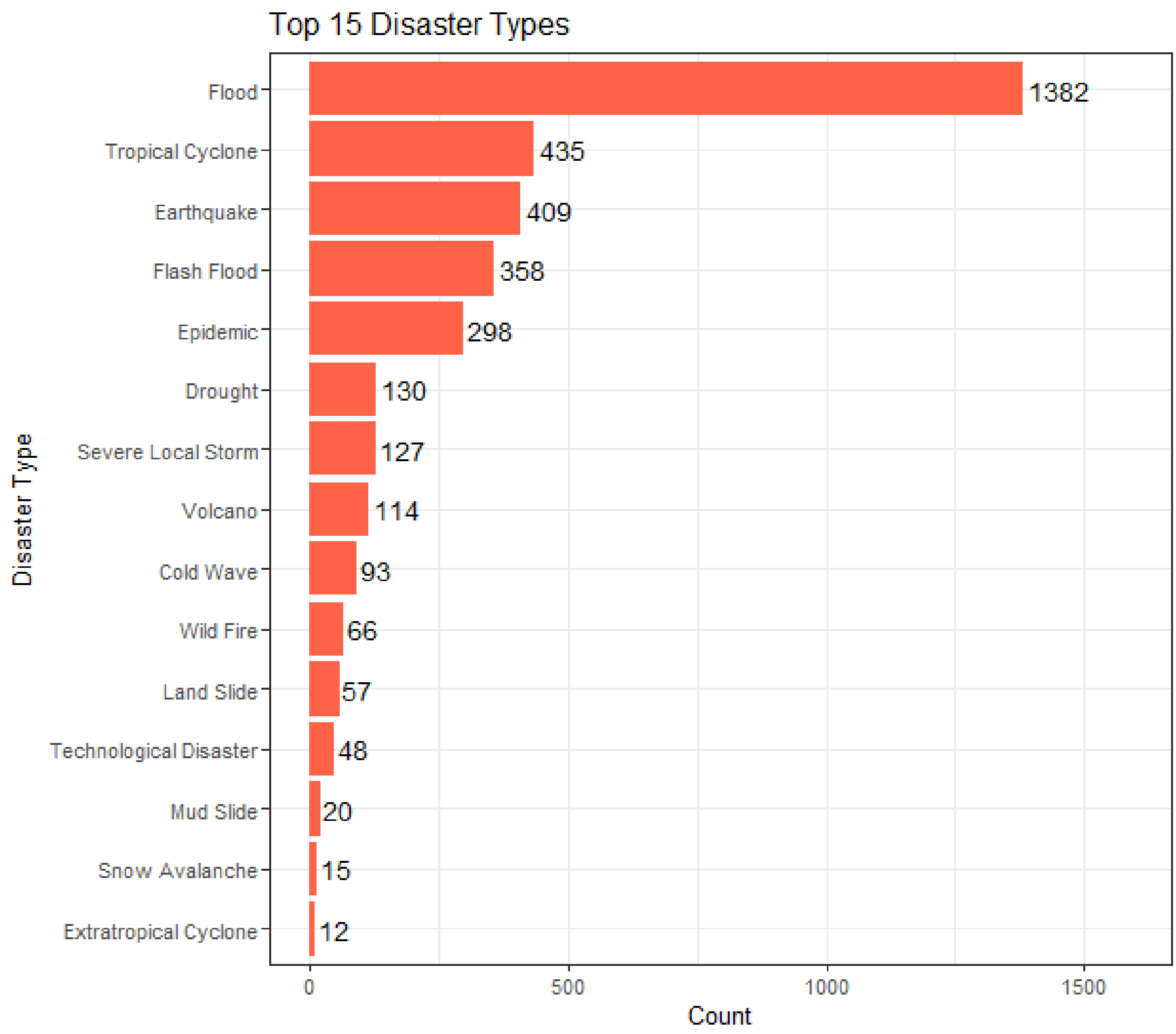
7. Exploratory Data Analysis (EDA)

EDA was conducted to explore distributions, relationships, and trends within the dataset.

7.1 Disaster Type Distribution

A bar chart was used to visualize the top 15 most frequent disaster types.

```r
# 8.1 Top 15 disaster types (bar)
top_types <- clean_disasters %>%
  count(DisasterType, sort = TRUE) %>%
  slice_head(n = 15)

ggplot(top_types, aes(x = reorder(DisasterType, n), y = n)) +
  geom_col(fill = "tomato") +
  geom_text(aes(label = n), hjust = -0.1, size = 4) +
  coord_flip() +
  expand_limits(y = max(top_types$n) * 1.15) +
  labs(title = "Top 15 Disaster Types", x = "Disaster Type", y = "Count")
```

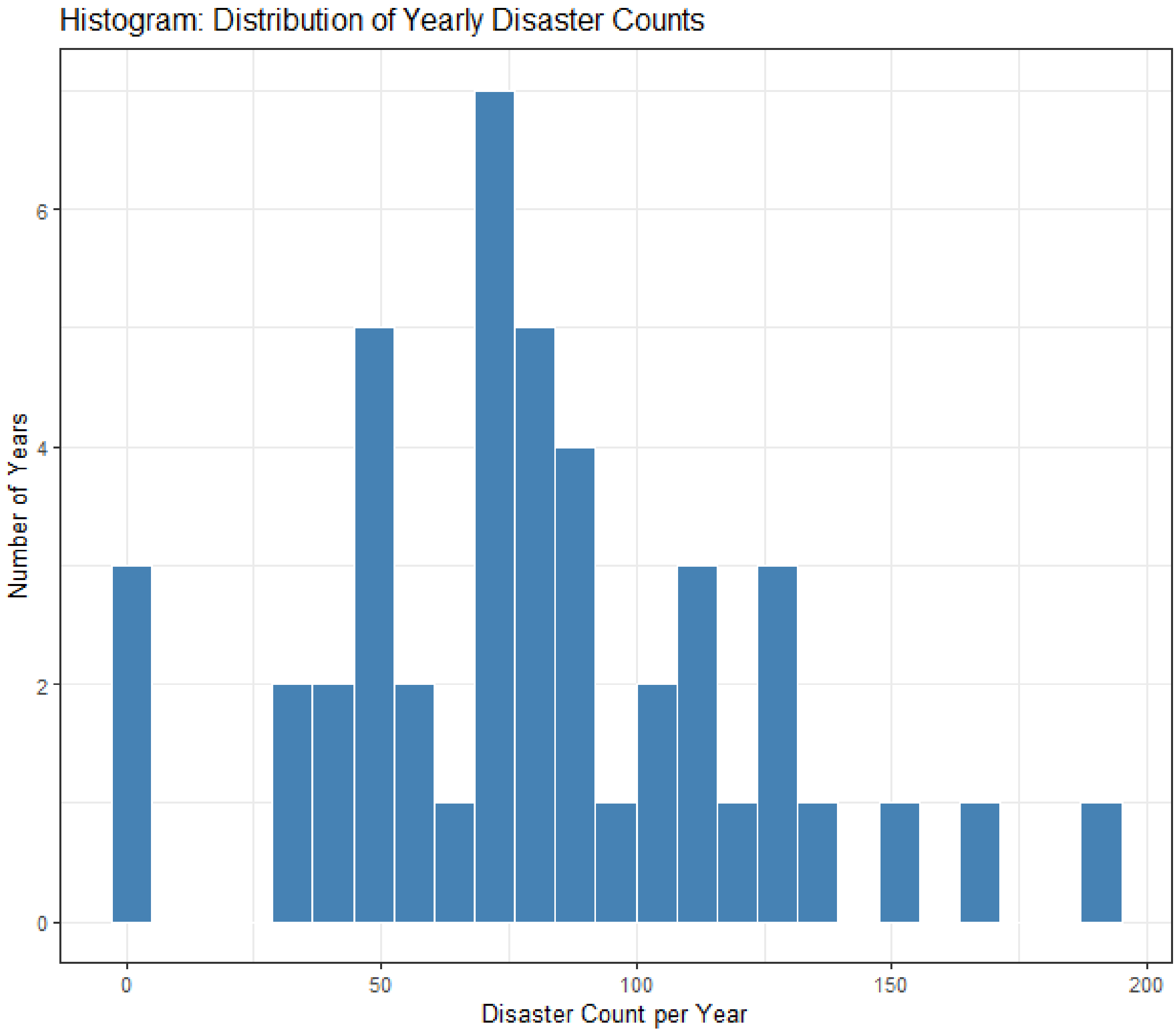


This bar chart shows the most frequently recorded disaster types in the dataset. Floods dominate the records, followed by tropical cyclones and earthquakes, indicating that climate-related disasters are the most common globally.

7.2 Temporal Distribution

A histogram of yearly disaster counts revealed a right-skewed distribution, indicating increasing disaster reporting over time.

```
# 8.2 Histogram (yearly counts distribution)
yearly_all <- clean_disasters %>%
  count(Year, name = "DisasterCount") %>%
  arrange(Year)

ggplot(yearly_all, aes(x = DisasterCount)) +
  geom_histogram(bins = 25, fill = "steelblue", color = "white") +
  labs(
    title = "Histogram: Distribution of Yearly Disaster Counts",
    x = "Disaster Count per Year",
    y = "Number of Years"
  )
```

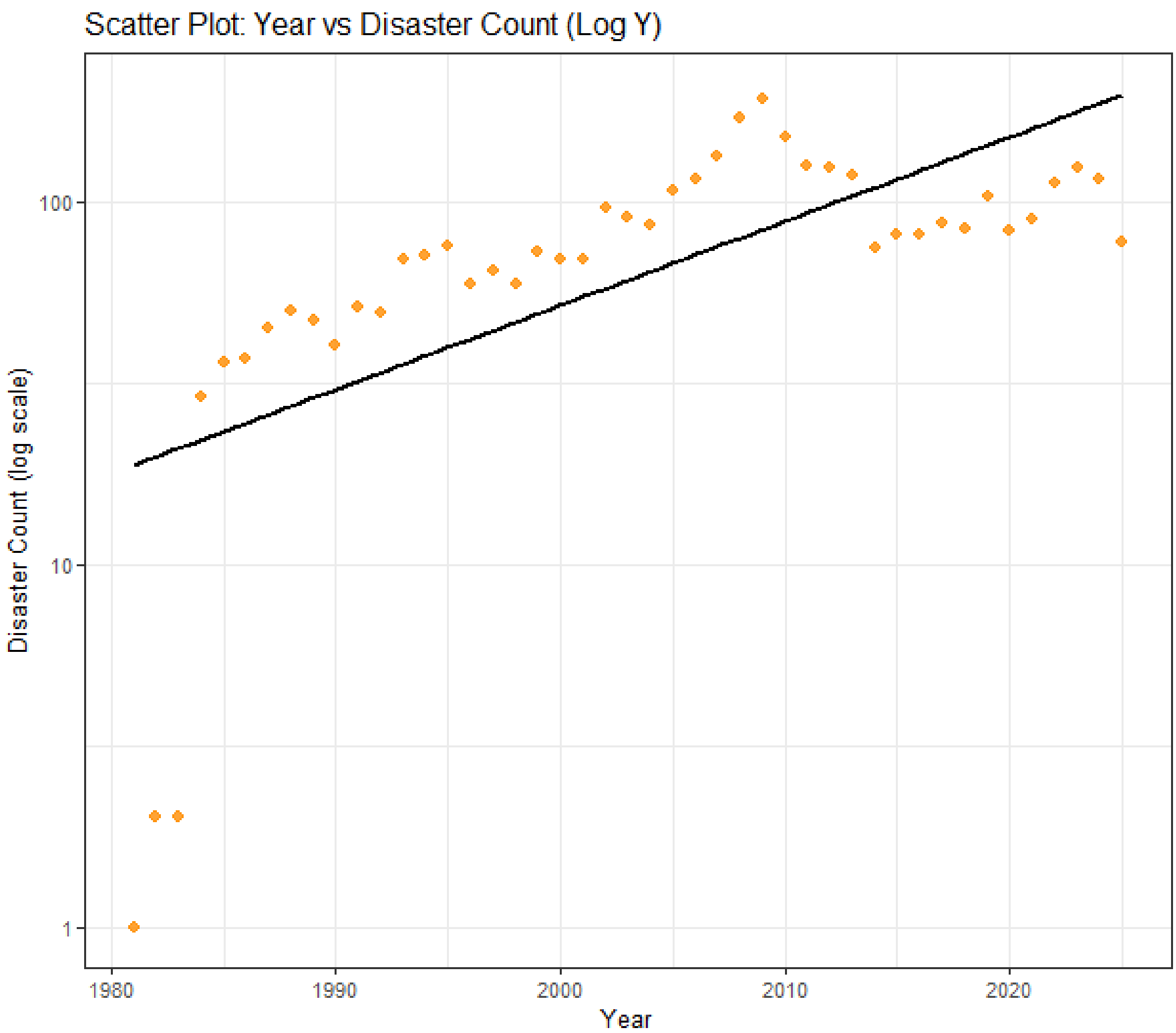


This histogram illustrates the distribution of the total number of disasters reported per year. The right-skewed distribution indicates that most years have moderate disaster counts, while a few recent years show significantly higher values.

7.3 Trend Analysis

A scatter plot with a log-scaled Y-axis illustrated the long-term trend in disaster occurrences.

```
# 8.3 Scatter (Year vs count) with trend, log Y
ggplot(yearly_all, aes(x = Year, y = DisasterCount)) +
  geom_point(color = "darkorange", size = 2, alpha = 0.8) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  scale_y_log10() +
  labs(
    title = "Scatter Plot: Year vs Disaster Count (Log Y)",
    x = "Year",
    y = "Disaster Count (log scale)"
  )
```

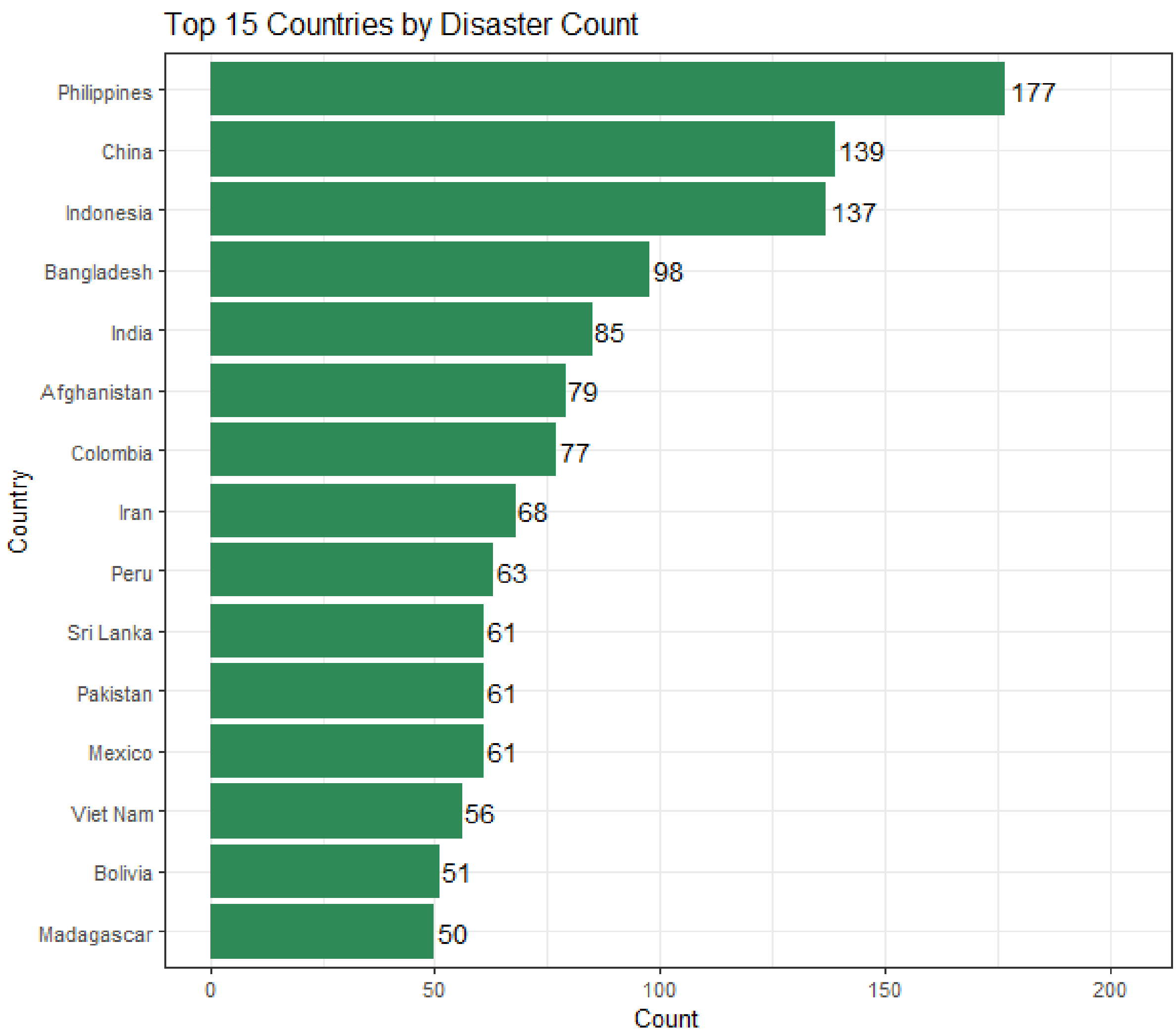


This scatter plot shows the relationship between year and disaster count using a logarithmic scale. The upward trend suggests an increase in reported disasters over time, while the log scale helps manage skewness caused by recent high-count years.

7.4 Country-wise Disaster Distribution

A bar chart highlighted the top 15 countries affected by disasters.

```
# 8.4 Top 15 countries (bar)
top_countries <- country_long %>%
  count(Country, sort = TRUE) %>%
  slice_head(n = 15)

ggplot(top_countries, aes(x = reorder(Country, n), y = n)) +
  geom_col(fill = "seagreen") +
  geom_text(aes(label = n), hjust = -0.1, size = 4) +
  coord_flip() +
  expand_limits(y = max(top_countries$n) * 1.15) +
  labs(title = "Top 15 Countries by Disaster Count", x = "Country", y = "Count")
```

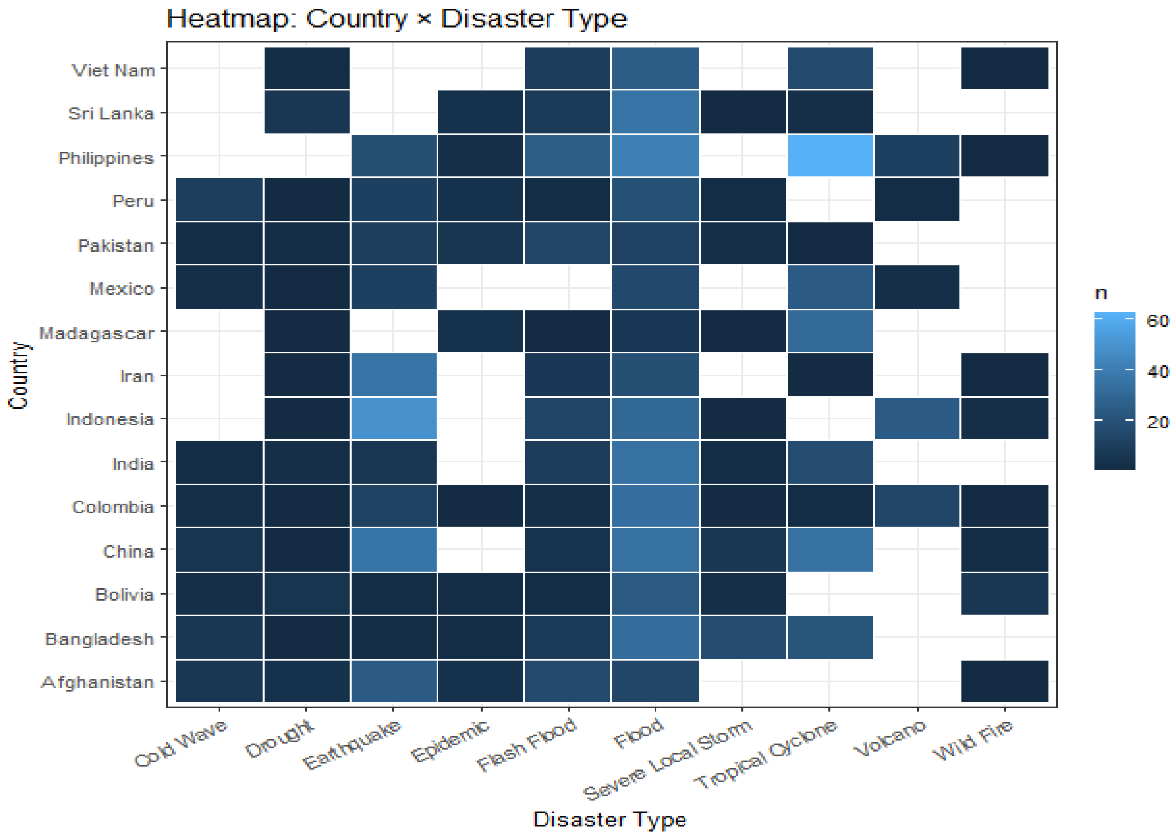


This chart highlights the countries most frequently affected by disasters. Countries such as the Philippines, China, and Indonesia appear most often, reflecting their geographic vulnerability to natural hazards.

7.5 Country × Disaster Type Relationship

A heatmap was used to analyze the relationship between top disaster-prone countries and disaster types.

```
# 8.5 Heatmap (top 15 countries x top 10 types)
top10_types <- clean_disasters %>%
  count(DisasterType, sort = TRUE) %>%
  slice_head(n = 10) %>%
  pull(DisasterType)

top15_country_names <- top_countries %>% pull(Country)

country_type <- clean_disasters %>%
  filter(DisasterType %in% top10_types) %>%
  separate_rows(CountriesRaw, sep = ",\\s*") %>%
  rename(Country = CountriesRaw) %>%
  mutate(Country = str_trim(Country)) %>%
  filter(Country %in% top15_country_names) %>%
  count(Country, DisasterType, name = "n")

ggplot(country_type, aes(x = DisasterType, y = Country, fill = n)) +
  geom_tile(color = "white") +
  labs(title = "Heatmap: Country x Disaster Type", x = "Disaster Type", y = "Country") +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```

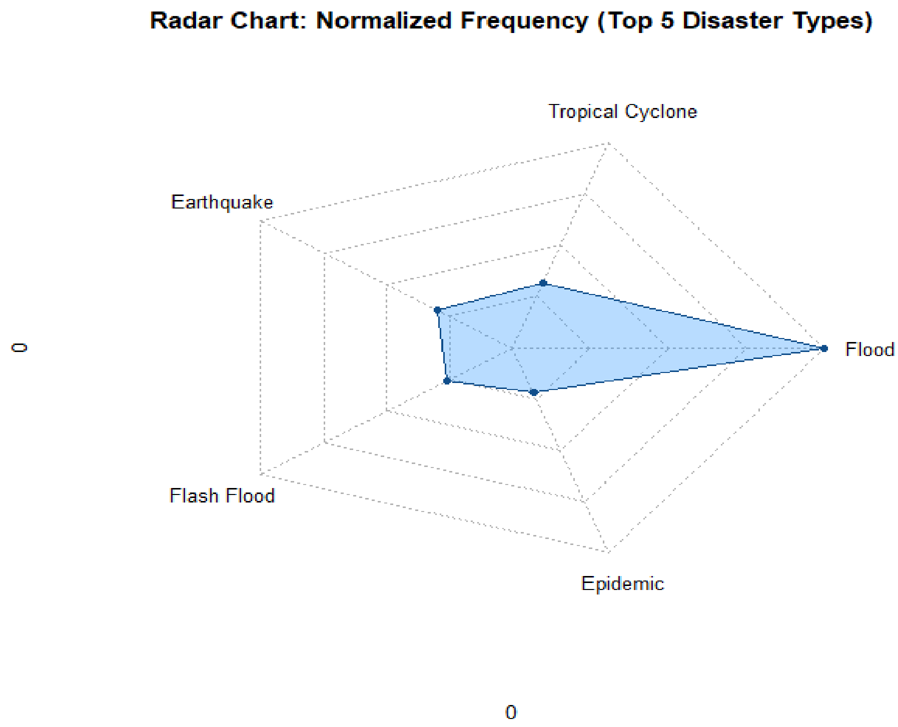


The heatmap visualizes the relationship between the top disaster-prone countries and major disaster types. Darker cells indicate higher frequencies, revealing country-specific disaster patterns (e.g., floods and cyclones in certain regions).

7.6 Relative Frequency Comparison

A radar chart visualized normalized frequencies of the top five disaster types.

```
# 8.6 Radar chart (top 5 types, normalized) - base R
top5 <- clean_disasters %>%
  count(DisasterType, sort = TRUE) %>%
  slice_head(n = 5)

radar_vals <- top5$n / max(top5$n)
radar_labels <- top5$DisasterType

radar_vals_closed <- c(radar_vals, radar_vals[1])
angles <- seq(0, 2 * pi, length.out = length(radar_vals_closed))

par(bg = "white")
plot(
  0, 0, type = "n",
  xlim = c(-1.2, 1.2), ylim = c(-1.2, 1.2),
  axes = FALSE,
  main = "Radar Chart: Normalized Frequency (Top 5 Disaster Types)"
)

for (r in seq(0.25, 1, by = 0.25)) {
  lines(r * cos(angles), r * sin(angles), lty = 3, col = "gray70")
}

axis_angles <- seq(0, 2 * pi, length.out = length(radar_vals) + 1)[-(length(radar_vals) + 1)]
for (i in seq_along(radar_labels)) {
  lines(c(0, cos(axis_angles[i])), c(0, sin(axis_angles[i])), lty = 3, col = "gray70")
  text(1.15 * cos(axis_angles[i]), 1.15 * sin(axis_angles[i]), radar_labels[i], cex = 0.9)
}

polygon(
  radar_vals_closed * cos(angles),
  radar_vals_closed * sin(angles),
  col = rgb(30, 144, 255, 80, maxColorValue = 255),
  border = "dodgerblue4"
)
points(radar_vals_closed * cos(angles), radar_vals_closed * sin(angles),
       pch = 16, cex = 0.8, col = "dodgerblue4")
```



This radar chart compares the relative frequency of the top five disaster types after normalization. It highlights the dominance of floods compared to other disaster types in a visually compact manner.

8. Classification Model

8.1 Problem Definition

The task was defined as a **multi-class classification problem**, where the goal was to predict the disaster category.

Target Variable:

- DisasterTypeClass (Top 7 disaster types + "Other")

Predictor Variables:

- Country
- Status Code
- Year

8.2 Model Selection

A **Random Forest classifier** implemented using the ranger library was selected due to its ability to handle:

- High-cardinality categorical variables
- Non-linear relationships
- Imbalanced real-world datasets

8.3 Model Performance

The trained model achieved:

- **Accuracy:** 43.29%

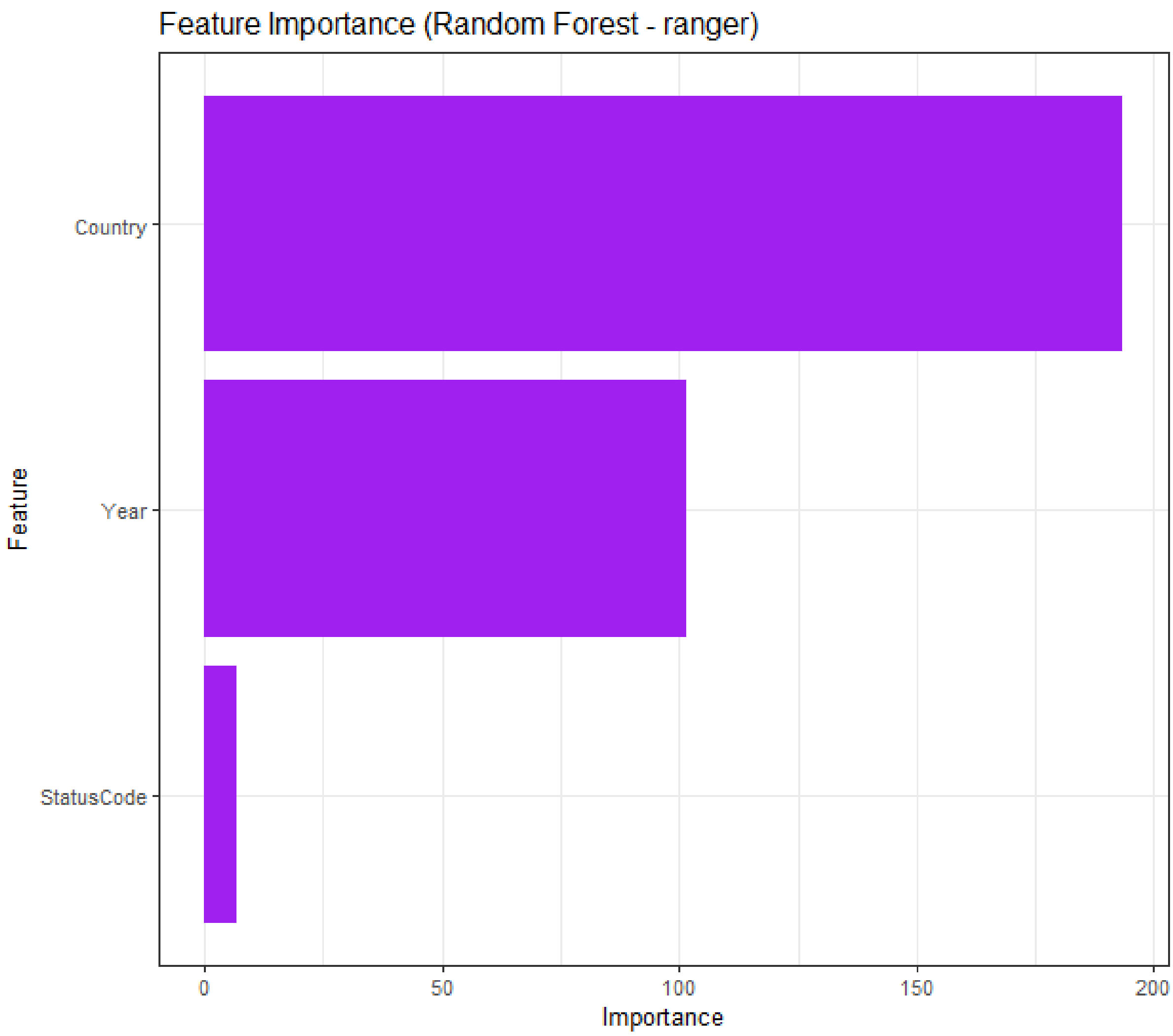
A confusion matrix heatmap was used to visualize classification performance, and feature importance analysis was conducted.

```r
# Part 9: Classification (Random Forest using ranger)
clf_data <- clean_disasters %>%
  separate_rows(CountriesRaw, sep = ",\\s*") %>%
  rename(Country = CountriesRaw) %>%
  mutate(Country = str_trim(Country)) %>%
  filter(Country != "")
# Top-N + Other to avoid too many rare labels
TOP_N <- 7
top_types_for_model <- clf_data %>%
  count(DisasterType, sort = TRUE) %>%
  slice_head(n = TOP_N) %>%
  pull(DisasterType)
clf_data <- clf_data %>%
  mutate(DisasterTypeClass = ifelse(DisasterType %in% top_types_for_model, DisasterType, "Other"))
model_df <- clf_data %>%
  select(DisasterTypeClass, Country, StatusCode, Year) %>%
  mutate(
    DisasterTypeClass = factor(DisasterTypeClass),
    Country = factor(Country),
    StatusCode = factor(StatusCode),
    Year = as.numeric(Year)
  )
cat("\nClass distribution (Top-N + Other):\n")
print(table(model_df$DisasterTypeClass))
# Train/test split
set.seed(42)
train_idx <- sample(seq_len(nrow(model_df)), size = 0.8 * nrow(model_df))
train_df <- model_df[train_idx, ]
test_df  <- model_df[-train_idx, ]
# Train Random Forest (ranger)
set.seed(42)
rf_ranger <- ranger(
  formula = DisasterTypeClass ~ Country + StatusCode + Year,
  data = train_df,
  num.trees = 500,
  importance = "impurity",
  respect.unordered.factors = "order",
  probability = TRUE
)
# Predict probabilities -> class
pred_prob <- predict(rf_ranger, data = test_df)$predictions
pred_class <- colnames(pred_prob)[max.col(pred_prob)]
pred_class <- factor(pred_class, levels = levels(test_df$DisasterTypeClass))
# Accuracy + Confusion Matrix
accuracy <- mean(pred_class == test_df$DisasterTypeClass)
cat("\nRanger Random Forest Accuracy:", round(accuracy, 4), "\n")
cat("\nConfusion Matrix:\n")
print(table(Predicted = pred_class, Actual = test_df$DisasterTypeClass))
# Feature importance (plot)
imp <- sort(rf_ranger$variable.importance, decreasing = TRUE)
imp_df <- data.frame(Feature = names(imp), Importance = as.numeric(imp))
ggplot(imp_df, aes(x = reorder(Feature, Importance), y = Importance)) +
  geom_col(fill = "purple") +
  coord_flip() +
  labs(title = "Feature Importance (Random Forest - ranger)", x = "Feature", y = "Importance")
```

Output:

```
Ranger Random Forest Accuracy: 0.4329
>
> cat("\nConfusion Matrix:\n")

Confusion Matrix:
> print(table(Predicted = pred_class, Actual = test_df$DisasterTypeClass))
                    Actual
Predicted             Drought Earthquake Epidemic Flash Flood Flood Other Severe Local Storm Tropical Cyclone
  Drought                   0          0        1           0     0     0                  0                0
  Earthquake                0          0        0           0     0     0                  0                0
  Epidemic                  1          0        3           1     1     0                  0                0
  Flash Flood               0          8        4          16     8     7                  1                0
  Flood                    20         65       46          59   257    71                 27               58
  Other                     0          0        0           0     2     2                  0                0
  Severe Local Storm        0          0        0           0     0     0                  0                0
  Tropical Cyclone          3         10        0           2    10     4                  1               35
>
```

```
Ranger Random Forest Accuracy: 0.4329
>
> cat("\nConfusion Matrix:\n")

Confusion Matrix:
> print(table(Predicted = pred_class, Actual = test_df$DisasterTypeClass))
                     Actual
Predicted             Drought Earthquake Epidemic Flash Flood Flood Other Severe Local Storm Tropical Cyclone
  Drought                   0          0        1           0     0     0                  0                0
  Earthquake                0          0        0           0     0     0                  0                0
  Epidemic                  1          0        3           1     1     0                  0                0
  Flash Flood               0          8        4          16     8     7                  1                0
  Flood                    20         65       46          59   257    71                 27               58
  Other                     0          0        0           0     2     2                  0                0
  Severe Local Storm        0          0        0           0     0     0                  0                0
  Tropical Cyclone          3         10        0           2    10     4                  1               35
>
```



This plot shows the relative importance of features used in the Random Forest model. Country is the most influential predictor, followed by year, indicating that geographic and temporal factors play a major role in disaster classification.

9. Model Evaluation and Discussion

Feature importance analysis revealed that:

1. **Country** is the most influential predictor
2. **Year** contributes moderately
3. **StatusCode** has limited predictive power

The moderate accuracy reflects class imbalance and the limited availability of explanatory features.

10. Limitations

- Disaster severity and impact metrics were unavailable
- Temporal reporting bias exists in recent years
- Rare disaster classes were underrepresented
- Geographic information was not grouped by region

11. Conclusion

This project demonstrates a complete data science workflow applied to real-world humanitarian data. Through web scraping, preprocessing, exploratory analysis, and classification modeling, meaningful insights were derived regarding global disaster patterns. Despite limitations, the study highlights the usefulness of data-driven approaches for humanitarian disaster analysis.

12. Future Work

- Incorporate disaster severity and damage estimates
- Group countries into regions for improved modeling
- Apply class weighting to improve minority-class prediction
- Extend analysis using natural language processing on disaster titles